

Dance Montage through Style Transfer and Music Generation

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Abstract

“Ink Dance” is an experimental video project exploring multimodal generation in dance montages. It aims to integrate dance, painting, and music using image style transfer and music generation, producing an aesthetic and innovative montage video where multiple art forms blend harmoniously. Specifically, we fuse Chinese dance with traditional ink painting and AI-generated music, creating a special form of dance expression called “Dance in the Painting”. This project demonstrates that multimodal generation techniques can introduce new presentation forms to classical art, offering fresh perspectives on the fusion of technology and art.

CCS Concepts

• Applied computing → Media arts.

Keywords

Dance Montage, Style Transfer, Music Generation

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1 Introduction

The development of multimodal generation has made significant strides in areas such as image generation [Oppenlaender 2022; Ramesh et al. 2022; Rombach et al. 2022; Zhang et al. 2023a] and music generation [Copet et al. 2024; Li et al. 2024a], which can be used to create digital paintings and music works. In the field of image generation, numerous studies have concentrated on neural style transfer [Gatys et al. 2016; Jing et al. 2020], which leverages convolutional neural networks (CNNs) [Deng et al. 2020; Zhang et al. 2023c], flow models [An et al. 2021], or Transformers [Deng et al. 2022] as encoders and decoders to disentangle and recombine image content and style features, thereby producing artistic images. Recently diffusion models [Rombach et al. 2022] have been employed to develop more advanced style transfer techniques, enabling the generation of high-quality and artistically sophisticated images [Deng et al. 2024; Zhang et al. 2023b]. In the field of music generation, significant research has explored the automatic production of digital music guided by video [Li et al. 2024a,b]. These studies leverage advancements in multimodal generation to synchronize

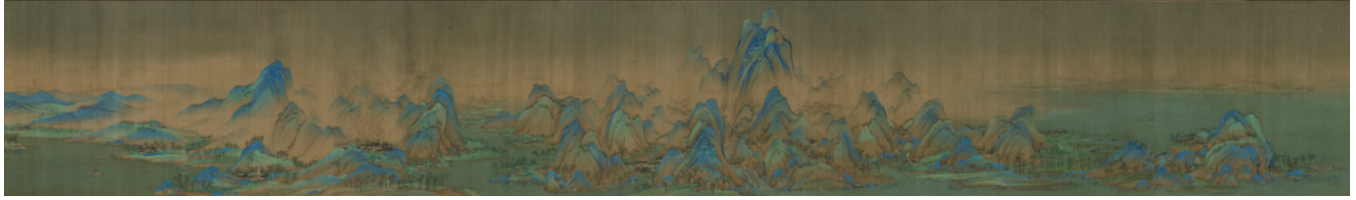


Figure 1: A Panorama of Mountains and Rivers (excerpt).



(a) Dance movement: Qinglv Waist



(b) Stylization of the movement

Figure 2: Transferring a traditional Chinese dance movement to the Chinese ink painting style can better express the artistic essence of dance movements.

audio and visual elements, creating cohesive and immersive multimedia experiences.

However, the potential of these techniques to empower other forms of art has not been extensively studied. This paper presents a montage video “Ink Dance” as an example to demonstrate that those advanced multimodal generation techniques can create innovative dance art forms through the concept of montage. Montage, in this context, refers to the technique of selecting, editing, and piecing together separate multi-model sections to form a continuous whole, which allows for a dynamic and engaging fusion of different artistic elements. Our work involves applying style transfer techniques to transform Chinese dance videos into the traditional Chinese ink painting style, with video-to-music generation enhancing the audiovisual experience. This research aims to bridge the gap between different art forms and offers fresh perspectives on the fusion of technology and traditional art.

2 Dance Montage

Our work, dance montage, is a brand new art form that combines tradition and modernity. In this project, we captured graceful movements of Chinese dancers and transformed them into vivid ink paintings, giving new life to classical art. Through the montage technique, we skillfully blend dance, painting, and traditional Chinese music to create a unified and harmonious artwork. The brushstrokes of the ink painting rise and fall with the rhythm of the

music, sometimes delicate and soft, sometimes unrestrained and powerful, which complements the dancers’ movements, together drawing a flowing musical picture.

The concept of montage is at the core of this work, as it emphasizes the technology of combining different multi-model elements into a unified whole, thereby achieving an innovative fusion of dance, painting, and music. Dance Video Montage provides viewers with a new way to experience and appreciate traditional art, which is praised for its aesthetics and cultural resonance. It emphasizes the potential of technology to not only preserve cultural heritage, but also innovate it.

3 Montage Video “Ink Dance”

3.1 Artistic Background

“A Panorama of Mountains and Rivers” is a masterpiece of Chinese art (see Figure 1 for an excerpt). Different from the black and white ink painting, this painting is named after the azurite green mineral pigment used in its creation. The gorgeous green tones are interwoven to depict the magnificent scenery of the vast rivers and mountains. The painting spans over 11 meters in length, allowing viewers to immerse themselves in a continuous journey through mountains, rivers, villages, and forests.

“Poetic Dance: The Journey of a Legendary Landscape Painting” is a Chinese dance performance inspired by this famous blue-green



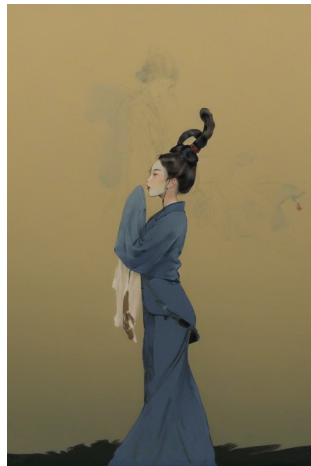
(a) Original dance movement



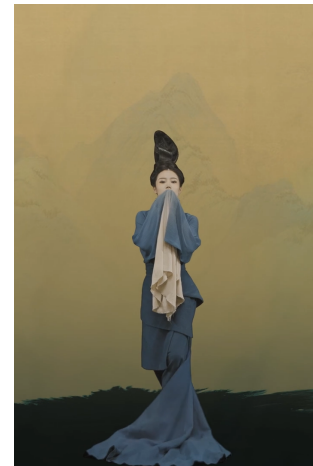
(b) Stylization result

Figure 3: Change a dance movement to ink painting style by style transfer (close-up of face).

(a) Original dance movement



(b) Stylization result

Figure 4: Change a dance movement to ink painting style by style transfer (side view).

(a) Original dance movement



(b) Stylization result

Figure 5: Change a dance movement to ink painting style by style transfer (front view).

landscape painting. The dance skillfully intertwines the visual elements of the painting with the fluid movements of traditional Chinese dance, creating a living representation of classical art. The famous movement “Qinglv Waist” draws its inspiration directly from the steep peaks in the landscape painting. The shape of the dancer’s waist is very similar to the contour of the mountain in the background (see Figure 2). This movement, performed with the dancer’s flexibility and strength, transforms the static peaks into dynamic artistic expressions, showcasing both the dancer’s skill and the essence of the painting.

3.2 Style Transfer

By stylizing the dance video into an ink painting, this project aims to deepen the connection between visual art and performance. To

achieve this artistic fusion, we employed an ink wash painting style transfer model¹ based on diffusion models [Rombach et al. 2022] and LoRA (Low-Rank Adaptation) [Hu et al. 2022] to stylize the dance video. The diffusion model iteratively optimizes and transforms dance video frames to reflect the style of ink painting, capturing delicate brushstrokes and complex textures. LoRA fine-tunes the pre-trained model to precisely control the style transfer process to achieve the desired artistic effect (see in Figures 3-5). Changes in ink color convey the rhythm and sense of space of the movements, creating dynamic harmony between the dance and the picture, emphasizing the combination of reality and illusion in traditional Chinese art.

¹<https://civitai.com/models/231966/ink-wash-painting>

Our goal is to create a unique combination of traditional art and modern performance through style transfer technology. The blue-green tones in the painting are carefully integrated into the video, enhancing the visual continuity between the original work and the dance. The texture and color gradient of the ink brushstrokes are simulated to give the video a realistic ink painting appearance. In this process, the dancer's movements seem to dance on the paper, and the ink color flows and diffuses with the changes in the movements, forming a dynamic painting with vitality. In this way, we not only preserve the essence of traditional Chinese art, but also give it new life and expression.

3.3 Music Generation

To generate background music for the montage video, we utilize a video to music generation method [Qian et al. 2024] to create background music that complements both the traditional Chinese dance style and ink painting style. The generated music integrates seamlessly with the visual elements, creating a cohesive and immersive audiovisual experience.

We first extract dance motion and scene information from 2D RGB frames. Based on this information, we then generate music that aligns both stylistically and rhythmically with the dance. The dance motion information includes the 3D coordinates of 33 body keypoints predicted by a human pose estimation algorithm, as well as a series of derived features such as limb-to-body center distances, joint angles, body contour dimensions, joint velocities, joint accelerations, joint pause detection, motion amplitude, and facial expression recognition. Scene information includes the background, objects in the frame, and the dancer's costume.

Using the above information, the music generation algorithm first calculates a series of musical meta-information. It employs a multi-layer perceptron model to predict style labels, which in turn generates a matching combination of instruments, scales, and chord progressions. Simultaneously, the algorithm selects the beat and initial tempo (bpm) that best synchronizes with the dance movements. These meta-information elements serve as global constants throughout the music generation process. For each frame, the algorithm calculates the current dance motion features and bpm, identifying significant limb movements and calculating their speed and amplitude. The model then predicts the next melody note under music theory constraints using the previously mentioned information. It combines the melody with chord textures under harmonic constraints and generates percussion accompaniment segments within the beat constraints. While generating music in real-time, the algorithm considers the movement features of each frame, producing synchronized melodies, chord accompaniments, and percussion to ensure the music perfectly aligns with the artistic styles of Chinese dance and ink painting in terms of style and rhythm.

4 Conclusion

This work presents a unique artistic expression method that merges traditional Chinese dance with Chinese ink painting and customized music through multimodal generation technology. The project "Ink Dance" shows the potential to innovate art forms, preserve and

transmit cultural heritage, demonstrating the charm of traditional Chinese art in contemporary contexts.

However, despite the achievements in implementing style transfer techniques across numerous dance and ink painting works, several artistic challenges have become apparent. Diffusion-based style transfer models sometimes struggle to faithfully reproduce the content of the source dance frames, leading to noticeable changes in the characters' movements, expressions, and costumes, thereby reducing frame-to-frame continuity. Additionally, training the style models often involves extensive parameter tuning. These challenges call for deeper and more innovative research in stylization in the future.

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